

HannariHamaguriHAN: An Agent Submitted to the ANAC 2026 Human-Agent Negotiation League

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Abstract

This report describes HANNARIHAMAGURIHAN, our agent for the ANAC 2026 Human-Agent Negotiation League (HAN). The agent is designed for bilateral human-agent negotiation where both outcome quality and communicative behavior matter. Its core negotiation behavior is deterministic and implemented as a BOA-style modular negotiator consisting of a behavior-aware adaptive offering policy, a conservative time-dependent acceptance policy, and a robust frequency-based opponent model. The latest version strengthens the use of observed partner behavior by classifying the opponent’s bidding pattern and by increasing the influence of the partner-utility estimate in the later stages of negotiation. Natural-language messages are generated from fixed templates rather than by calling a large language model during negotiation. This design keeps the agent predictable, low-latency, and robust under tournament conditions, while still providing concise and cooperative messages to the human negotiator.

1 Introduction

The Human-Agent Negotiation League (HAN) differs from purely agent-agent negotiation settings because the agent must negotiate with people, not only with automated opponents. A competitive HAN agent should therefore satisfy two requirements at the same time. First, it must protect its own utility and avoid agreements below the reservation value. Second, it must behave in a way that is understandable to humans, avoiding unexplained jumps, excessive repetition, and unnecessarily aggressive language.

HANNARIHAMAGURIHAN follows this principle by separating strategic decision making from language generation. The outcome-level strategy is computed by a deterministic BOA architecture. The text layer then explains the strategic move using short templates, such as starting with a clear proposal, acknowledging the partner’s proposal, moving toward a practical compromise near the deadline, or accepting a workable agreement. Although the constructor accepts LLM-related arguments for compatibility with the HAN template, these arguments are not used during tournament execution. This avoids the latency and instability that may arise from online LLM calls while preserving human-readable negotiation behavior.

The main goals of the agent are:

- to maintain high utility early in the negotiation;
- to concede in a controlled way as time pressure increases;
- to classify observed partner behavior when enough evidence is available;
- to use the opponent model more strongly in late-stage candidate selection;
- to avoid accepting outcomes below the reservation value;
- to reduce offer repetition and choose interpretable counter-offers;
- to communicate in a cooperative but concise style.

2 The Design of HannariHamaguriHAN

The agent is implemented as a meta-negotiator named HannariHamaguriHAN. Internally, it wraps one BOA negotiator with three components:

1. **Offering policy:** `AdaptiveOfferingPolicy`, which chooses counter-offers from the feasible outcome space.
2. **Acceptance policy:** `AdaptiveAcceptancePolicy`, which decides whether to accept the current offer.
3. **Opponent model:** `RobustFrequencyOpponentModel`, which combines two frequency-based models.

All components share the same utility function and negotiation meta-information. At the start of a negotiation, the offering policy enumerates or samples the outcome space, filters out outcomes below the reservation

value, normalizes utilities, and sorts the remaining outcomes from best to worst. This preprocessing enables fast online decisions during the negotiation.

In the latest implementation, the default offering policy is initialized with tuned parameters rather than relying only on the class defaults. Partner influence starts earlier, at $t = 0.58$, and its maximum strength is increased to 0.76. The confidence cap for the partner model is also increased to 0.55. In the final phase, the partner-utility floor is set to 0.82, so among acceptable final candidates the agent favors offers that are also likely to be acceptable to the partner. The hardliner-specific settings are also adjusted: the late target discount is 0.05, the hardliner late floor is 0.27, and the hardliner final floor is 0.08.

The meta-negotiator returns `ExtendedOutcome` and `ExtendedResponseType` objects when appropriate. These objects attach a short text message to the strategic action. For example, the first proposal is accompanied by a message that frames the offer as a practical basis for discussion. Middle-stage counter-offers state that the agent can move from its earlier position while keeping the agreement workable. Late-stage offers are described as practical compromises. Acceptances and ending messages are also produced from deterministic templates.

A small implementation detail is that the agent observes the current offer before proposing or responding. The observed offer is passed to both the opponent model and the offering policy, while duplicate observations within the same step are ignored. This makes the strategic components aware of the partner’s latest behavior without double-counting the same offer.

3 Concealing Bidding Strategy

The bidding strategy is based on a target utility curve over relative time. Let $t \in [0, 1]$ be the relative negotiation time. The implementation uses three main utility targets: an early target of 0.9828, a middle target of 0.84, and a late target of 0.46. For $t < 0.30$, the agent remains highly ambitious. Between 0.30 and 0.75, it linearly moves from the early target toward the middle target. After 0.75, the concession curve becomes steeper and approaches the late target using a square-root time effect. This produces strong early self-protection and more visible compromise close to the deadline.

The latest version adds an explicit behavior-aware adjustment to this target curve. After at least four received offers, the offering policy classifies the partner into one of five behavioral types: **hardliner**, **conceder**, **random**, **stubborn**, or **adaptive**. The classification uses the ratio of unique offers, the movement between the first and last received utilities, and recent utility volatility. A low unique-offer ratio below 0.35 is treated as **hardliner**. A utility movement greater than 0.12 is treated as **conceder**. Recent volatility above 0.25 is treated as **random**. Almost no movement, below 0.04 in absolute value, is treated as **stubborn**. If no special pattern is detected, the partner is treated as **adaptive**; with fewer than four observations, the type is **unknown**.

This classification directly changes the target used for candidate generation. If the partner is a **conceder** and the negotiation has not yet reached $t = 0.85$, the agent increases its target by 0.02 and waits for further concessions. If the partner is a **hardliner** or **stubborn** after $t = 0.85$, the agent lowers its target by 0.05 in the submitted default configuration. If the partner is **random**, the agent increases the target by 0.02, avoiding excessive concessions to unstable behavior. These adjustments make the policy more reactive than the previous version, where the opponent type was effectively unused in the main offering decision.

The strategy conceals its exact bidding policy from the human counterpart in two ways. First, it does not reveal numerical utility thresholds in natural language. Its messages only express qualitative movement, such as “I can move from my earlier position” or “I am moving to a practical compromise.” Second, the offering policy avoids immediately repeating the same offer. If the intended offer is identical to the most recently sent offer, the policy searches for a nearby alternative with comparable utility. This makes the behavior appear less mechanical while preserving control over the agent’s own utility.

The offering policy also reacts to the partner’s history independently of the behavior class. It stores received offers and their normalized utilities. If the partner’s offers improve over time, the agent makes a small additional concession late in the negotiation. If the partner’s offers are nearly stationary, the agent lowers its target after the middle phase to increase the chance of agreement. Near the deadline, the agent may replay the best received offer if it is sufficiently good. Specifically, after $t = 0.93$, a received offer with normalized utility at least 0.90 can be replayed; after $t = 0.975$, the replay threshold is relaxed to 0.25. If the best received offer is weak but provides useful orientation, the policy can instead search near that offer for a small improvement that is acceptable to the agent.

For **hardliner** and **stubborn** partners, the latest version includes a special late-stage branch after $t = 0.94$. In that branch, the candidate floor becomes $\max(0.27, \text{target} - 0.14)$ until $t = 0.96$, and then becomes 0.08 after $t = 0.96$. The selected offer is chosen to be close to the partner’s reference offer, with partner score and own utility used as tie-breakers. This branch is intended to recover possible agreements with non-conceding partners while still keeping a minimum utility floor.

The final and pre-final stages are also expressed through named policy parameters. After $t = 0.96$, the pre-final candidate floor is 0.48, the own-utility slack is 0.30, and the policy prefers offers close to a compromise

target of 0.50. After $t = 0.98$, the final minimum floor is 0.02, but replaying the partner’s exact reference offer requires at least 0.35 normalized utility. The final own-utility band also uses 0.35 as its floor and 0.40 as its slack, while the final partner floor is tuned to 0.82 in the default agent. These values make the deadline behavior more configurable and easier to reason about than direct constants inside the decision logic.

Candidate selection balances three criteria:

- **own utility**, using normalized utility above the current target or phase-specific floor;
- **estimated partner utility**, using the opponent model after enough observations;
- **distance from the partner’s latest or best received offer**, so counter-offers remain interpretable.

For numeric outcome spaces, distance is normalized issue by issue when the issue ranges are available. For general tuple outcomes, distance is computed as the fraction of differing issue values. This prevents the agent from making arbitrary-looking jumps when a nearby compromise is available.

The implementation also includes several domain-aware safeguards. For four-integer domains and eight-categorical domains, the late rescue slack is disabled, so the agent does not over-relax its offer floor in domains where this was judged risky. For other domains, a rescue slack can gradually increase from $t = 0.86$ to $t = 0.96$, with a maximum value of 0.12. This allows more flexible late concessions in suitable domains while preserving stricter behavior in selected domain types.

4 Acceptance Strategy

The acceptance policy is intentionally conservative. It never accepts an offer whose utility is below the reservation value. Accepted offers must first pass this absolute safety check, and only then are they compared with normalized time-dependent thresholds.

The policy uses nominal early, middle, and late threshold parameters of 0.98, 0.94, and 0.97. These thresholds are further protected by a reservation margin of 0.02 and by an ordering rule that keeps earlier phases at least slightly stricter than later phases. In ordinary rounds, this makes acceptance difficult unless the offer is very strong. In large-scale utility domains, the implementation adds an extra late-stage strictness adjustment after $t = 0.75$ when the utility scale is greater than 2.0.

The policy contains two deadline-oriented mechanisms. First, after relative time 0.95, offers with normalized utility at least 0.75 are accepted to avoid losing good agreements near the end. Second, at $t \geq 0.99$, the final threshold is relaxed to 0.10, but still only after the absolute reservation-value check has been passed. This final rule reduces negotiation failure when time is almost exhausted while still preventing agreements below the reservation value.

The following table summarizes the main acceptance parameters.

Parameter	Role
Early / middle / late thresholds	Main nominal time-dependent acceptance parameters: 0.98, 0.94, and 0.97.
Urgent acceptance time and threshold	After $t = 0.95$, accepts offers with normalized utility at least 0.75.
Final threshold	At $t \geq 0.99$, accepts low but still reservation-safe offers to avoid failure.
Reserved margin	Keeps acceptance behavior safely above the reservation value.
Large-scale adjustment	After $t = 0.75$, increases strictness when the utility scale is greater than 2.0.

5 Opponent Model

The opponent model, `RobustFrequencyOpponentModel`, is a softened frequency model. It combines the Hard-Headed frequency model with the Smith frequency model. If $H(o)$ is the HardHeaded score for outcome o and $S(o)$ is the Smith score, the blended score is computed as a geometric mixture:

$$U_{opp}(o) = H(o)^{1-w} S(o)^w,$$

where $w = 0.1126$ in the implementation. A small epsilon is added before blending to avoid zero-score collapse.

The purpose of this model is not to perfectly reconstruct the human’s utility function. Instead, it gives a lightweight estimate of which outcomes are likely to be acceptable to the partner based on repeated issue values and observed proposals. This estimate is used mainly after several received offers have been observed. In the submitted default configuration, the offering policy limits model confidence with a cap of 0.55, so early behavior is still dominated by the agent’s own utility. As time advances, partner utility receives increasing influence through a joint score. Partner influence starts after $t = 0.58$ and is capped at 0.76.

The final phase uses the opponent model even more selectively. After $t = 0.98$, candidate offers are filtered through a partner floor of 0.82 or, when the candidate set is weaker, through a floor based on 85% of the best partner score in the candidate set. This makes the agent more agreement-oriented at the deadline without allowing the opponent model to override the reservation-value and own-utility constraints.

The public opponent utility property also adapts the blend depending on the negotiation domain. For example, the implementation uses special handling for four-integer outcomes and eight-categorical outcomes, reflecting that different domain structures may favor different frequency signals. In four-integer domains, the public blend strongly incorporates inverse own utility together with the robust frequency estimate. In eight-categorical domains, it relies mainly on the frequency signal. This domain-aware blending is a pragmatic technique to improve robustness across heterogeneous HAN scenarios.

6 Evaluation

The current version was evaluated from four perspectives: strategic safety, concession behavior, opponent orientation, and human-facing communication.

Strategic safety. The most important safety property is that the acceptance policy checks the reservation value before applying any normalized threshold. Therefore, even the final relaxed acceptance rule cannot accept an outcome below the reservation value. The offering policy also filters candidate outcomes below the reservation value during preprocessing.

Concession behavior. The target curve starts near the maximum utility and decreases gradually. This allows the agent to protect utility in the early phase, make controlled concessions in the middle phase, and become substantially more agreement-seeking near the deadline. The latest version further adapts this curve according to observed partner behavior. It can wait longer against conceding partners, become more flexible against hardliners or stubborn partners late in the negotiation, and avoid over-reacting to random-looking proposals. Additional late flexibility is provided by replaying good received offers, searching near weak received offers for acceptable improvements, and applying rescue slack in selected domain types.

Opponent orientation. After enough observations, candidate offers are scored by a mixture of own utility and estimated partner utility. The updated default parameters make this orientation stronger than before: model confidence is capped at 0.55, partner influence begins at $t = 0.58$, and the influence cap is 0.76. The agent also tries to choose offers close to the latest or best received offer when this can be done without sacrificing too much utility. This is especially important in HAN because humans may interpret nearby counter-offers as constructive and large unexplained jumps as uncooperative.

Communication. The language layer was designed to be short, polite, and stable. The agent does not make misleading claims about its private utility. Instead, it provides high-level explanations of its move, such as starting with a clear proposal, acknowledging the partner’s proposal, moving from an earlier position, or accepting a workable balance. This avoids hallucinated arguments and keeps response latency predictable.

A limitation of the current evaluation is that it does not yet include a large-scale human-subject study. Therefore, future testing should compare this template-based communication style against richer LLM-generated messages, measuring not only utility and agreement rate but also perceived fairness, transparency, and user satisfaction.

7 Lessons and Suggestions

The main lesson from the design of HANNARIHAMAGURIHAN is that HAN agents should not rely only on language fluency. Human-facing negotiation still requires a stable strategic core. A persuasive message is harmful if it accompanies erratic offers, unexplained concessions, or reservation-violating acceptance. For this reason, our agent prioritizes deterministic strategy and uses language only to explain actions at a high level.

A second lesson is that avoiding repetition matters. Humans may see repeated offers as stubbornness or system failure. The offering policy therefore records recently sent offers and selects nearby alternatives when possible. This small behavioral detail can make the negotiation feel more natural without changing the strategic target.

A third lesson is that opponent modeling should affect both candidate scoring and high-level concession control. In the latest version, the opponent model is still bounded, but its influence starts earlier and reaches a higher cap. In addition, the policy classifies the observed bidding pattern and changes its target depending on whether the partner appears to be conceding, hardlining, stubborn, random, or adaptive. This makes the agent more responsive while preserving deterministic and reservation-safe behavior.

For future work, we plan to improve the human-facing text layer while keeping the deterministic fallback. One possible extension is to use an LLM only when enough time is available, with a strict timeout and a template fallback. Another extension is to detect negative or confused human messages and adjust the tone of the

explanation without changing the strategic constraints. Finally, human-subject experiments are needed to determine whether nearby counter-offers, behavior-aware concessions, and concise explanations improve perceived fairness.

Conclusions

HANNARIHAMAGURIHAN is a deterministic, low-latency HAN agent built around a BOA negotiation architecture. Its offering policy uses a time-dependent target curve, behavior-aware target adjustment, opponent-oriented candidate selection, replay of strong received offers, near-received-offer improvement, special late handling for hardliner and stubborn partners, and anti-repetition mechanisms. Its acceptance policy is reservation-safe and becomes more agreement-seeking only near the deadline. Its opponent model blends HardHeaded and Smith frequency estimates to provide robust but bounded partner-utility guidance, with stronger late-stage influence in the submitted default configuration. The agent’s natural language is template-based, concise, and cooperative, making it suitable for HAN settings where strategic reliability and human interpretability are both important.